

Remote digital health technologies for improving the care of people with respiratory disorders



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Respiratory diseases are a leading cause of morbidity and mortality globally. However, existing systems of care, built around scheduled appointments, are not well designed to support the needs of people with chronic and acute respiratory conditions that can change rapidly and unexpectedly. Home-based and personal digital health technologies (DHTs) allow implementation of new models of care catering to the unique needs of individuals. The high number of respiratory triggers and unique responses to them require a personalised solution for each patient. The real-world, repetitive monitoring capabilities of DHTs enable identification of the normal operating characteristics for each individual and, therefore, recognition of the earliest deviations from that state. However, despite this potential, the number of clinical efficacy studies of DHTs is quite small. Evaluation of clinical effectiveness of DHTs in improving health quality in real-world settings is urgently needed.

Introduction

Respiratory conditions are a tremendous burden on the individuals who have them, their families, health-care providers caring for them, and society as a whole.¹ Digital health technologies (DHTs) could lessen this burden, but there is no consensus on what variables to record to measure the relevant clinical status and changes for people with respiratory conditions.² In this Viewpoint, we provide example cases of how the application of remote sensor data and other personal DHTs might be incorporated into patient care to monitor people with chronic respiratory conditions remotely and allow early detection of acute respiratory infection. These examples allow us to describe both the potential benefits and remaining key gaps regarding DHT use in these and other cases to improve the care of people with respiratory conditions.

Lynnae, a fictional character, is a woman aged 37 years, with approximately a 20-year history of difficult-to-control asthma with recurrent exacerbations that require hospitalisation three to four times per year. Lynnae lives in rural Alaska, USA, and receives most of her medical care through a nurse practitioner in her hometown, supplemented by primarily virtual follow-up with a regional pulmonologist, who is otherwise reachable only by plane. Between routine medical appointments, the pulmonologist monitors her rescue inhaler use through a connected, digital inhaler. Shortly after starting to use the smart inhaler about 2 years ago, Lynnae and her pulmonologist recognised that her use of the rescue inhaler increased substantially during periods of poor air quality (figure 1). In response, Lynnae installed high-efficiency particulate air filters in her home and started tracking the indoor air quality with a monitor.

Lynnae's case highlights both the challenges that individuals with severe chronic respiratory diseases face and some of the potential advantages of incorporating remote DHTs into their care. For many people with chronic respiratory conditions, the substantial burden of self-management and constant hypervigilance for environmental triggers and potential infectious risks

negatively affects their quality of life. Too often, patients with respiratory illness can feel they bear the responsibility for the episodic deteriorations in their condition.³

Existing systems of care, built around somewhat arbitrarily timed routine follow-up visits, with the emergency room often as a sole contingency option, are not well designed to support the needs of people with chronic respiratory diseases, including Lynnae, whose condition can change rapidly and unexpectedly. People at risk of respiratory decompensation—rapid worsening in their ability to breathe—have few options for rapid evaluation when they begin to have early symptoms of allergies or infections, leading to delays in diagnosis, isolation, and treatment.⁴

Fortunately, many existing gaps in the health-care continuum are now well suited to be addressed through the implementation of redesigned systems of care built around the continuously increasing variety of novel home-based and personal DHTs. The term DHTs is still quite novel and rapidly changing. In this Viewpoint, we define DHTs based on previously published criteria that include technologies that: collect clinical or health-related data; contain both a physiological sensor and embedded software; are portable (not restricted to facility use); connect to the internet or other internet-connected technologies to transmit data; and are designed for patient-facing use (minimal clinician involvement).⁵

Burden of respiratory conditions

The lungs are the only internal organs continuously exposed to our environment, including its toxins, allergens, particulate matter, and infectious agents. Given the diversity and constancy of insults to the airways, the fact that respiratory diseases are a leading cause of death⁶ is not surprising. Although there are numerous chronic respiratory conditions, including cystic fibrosis, interstitial lung disease, and many more, the two most common chronic respiratory diseases are asthma and chronic obstructive pulmonary disease (COPD), which will be the primary focus of this Viewpoint. Lower respiratory tract infections, which are

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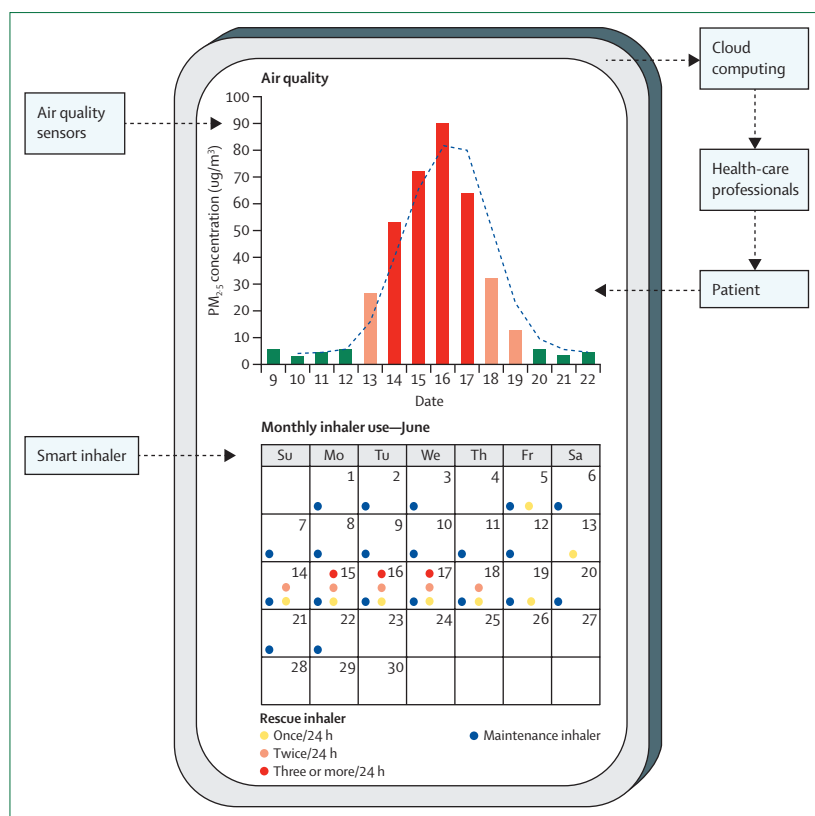


Figure 1: Multiple sensor data streams, such as automated tracking of inhaler use frequency, and indoor and outdoor air quality, can be combined to better inform the user and their care provider of personal exacerbating factors

the leading cause of infectious deaths annually, especially so in children younger than 5 years and adults older than 70 years, will also be addressed.⁷ With roughly 7 million deaths attributed to COVID-19 at the time of this writing, most due to respiratory complications, the effect of respiratory infections is especially evident.⁸

The true burden of respiratory diseases extends beyond excess mortality and morbidity, exacting a substantial emotional and financial cost to the affected individuals, their families, and societies. Beyond mortality, asthma is the leading cause of missed school days for children, affecting not only their educational attainment but also increasing stress and lowering the quality of life of their families and caregivers.⁹ The majority of people with COPD find their symptom burden to be a substantial challenge to their routine daily activities and, therefore, their quality of life.¹⁰ Although the short duration of symptoms for most people with an acute respiratory infection creates a minimal individual burden, the high incidence rate of these infections creates an enormous societal economic burden. As staggering as the estimated US\$11.2 billion average cost of annual influenza is in the USA, that number was transcended by the \$16 trillion estimated cost of the COVID-19 pandemic in 2020.^{11,12}

The problems to be solved

All people are affected by respiratory illness. For most, addressing the sporadic and unpredictable nature of acute infection might be all that is needed. But for more than half a billion people with chronic respiratory conditions, like Lynnae, and those close to them, the challenges to be addressed are more ubiquitous and complex.¹³ Additionally, as populations worldwide age, the incidence of chronic pulmonary conditions will continue to increase.¹⁴ This change will require already struggling health-care systems to adapt to better meet the needs of populations with chronic and acute recurring respiratory conditions, necessitating reorganisation of systems of care to enable informed self-management of chronic and acute therapeutic changes. DHTs, when implemented in an innovative and person-centric manner, can provide the flexibility of design and scale needed to better address the unique needs of each individual. DHTs also might reduce inequities in access to health care for underserved populations¹⁵ and lessen the need for unnecessary travel and other resource use.

Although the unmet needs are numerous, the overarching goal of any solution is to provide the individual patient with a personalised condition management strategy imposing the lowest burden possible. What works best for the unique needs of Lynnae would not necessarily be helpful for others. A successful approach must enable the user to maximise their functional capacity while minimising life-disrupting management activities and, most importantly, acute exacerbations and their consequences.

A distinctive capability of personal sensor technologies that helps make this personalised care possible is their ability to identify the normal operating characteristics for each individual and early deviations from these. In Lynnae's case, that was a change in her rescue inhaler use. In future scenarios, actionable, personalised, monitored parameters could include unexpected changes in heart or respiratory rate during routine activities.

The individual nature of respiratory triggers and a person's unique response to them mandates a personalised solution to substantially improve health. As with Lynnae, environmental triggers can be learned through the right sensors and collected data and, once identified, can be used to proactively minimise exacerbations. Similarly, public health data, along with physiological data, can inform avoidance or early diagnosis and treatment strategies. As neighbourhoods exposed to high concentrations of air pollutants are often low income, the combination of personal and public data could be especially valuable in beginning to address this aspect of health inequities.¹⁶

Addressing chronic respiratory illnesses

The selection of novel DHTs capable of assisting in respiratory care is vast and growing (figure 2; table 1). However, irrespective of the inventiveness of a technology,

DHT is rarely a standalone solution in patient care. Rather, DHTs need to be considered as a novel tool with the use of which inventive approaches to care can be built—approaches that can motivate and encourage users to feel comfortable engaging in their own care.

Personal physiological sensors, such as wrist-worn wearables, are a key technology for the development of solutions that can improve chronic conditions management, as they enable care determinations to be based on

personal, rather than population, ranges of normal values. Although the majority of wrist-worn wearables currently available are consumer devices with minimal to no regulatory review of the physiological and behavioural metrics they track, newer devices, such as Corsano (Switzerland), Empatica (USA), and Biobeat (Israel) are examples of multiparametric wrist-worn wearables that have obtained or are seeking regulatory approval in the USA, Europe, and beyond for a range of vital parameters, including pulse, blood pressure, and oxygen saturation. All of these wearables continuously track a range of vital signs and activity, and have been found to allow for the early detection of respiratory infections.^{17,18}

For individuals with chronic respiratory conditions, daily use of these sensors can allow for recognition of changes in their vital parameters, such as increases in nocturnal respiratory rate, which can possibly herald exacerbations.¹⁹ Aggregated continuous data collection might increase our understanding of variabilities in respiratory diseases and contribute to defining increasingly refined phenotypes.

Beyond wearable physiological sensors, multiple additional technologies can play an important role in the creation of digital novel approaches to improving the care for people with respiratory illness. Smart inhalers are digital monitoring systems connected to inhalers that can instantaneously transmit an array of inhaler usage information to a connected device in a user-friendly

For more on **Corsano** see <https://corsano.com>

For more on **Empatica** see <https://www.empatica.com>

For more on **Biobeat** see <https://www.bio-beat.com>

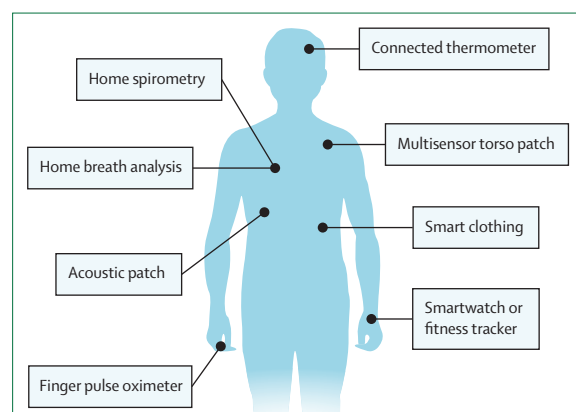


Figure 2: Available personal sensor technologies that have been, or could be, used within redesigned systems of care built to better identify early signs of respiratory decompensation and lessen the burden of care for patients

	Parameters measured*	Continuous or intermittent tracking	Potential clinical use	Device FDA regulatory clearance†	User's role‡	Clinician's or health-care system's role‡
Smartwatches, fitness trackers, smart rings, and smart clothing	Activity, pulse rate and irregular rhythm detection, heart rate variability, respiratory rate, oxygen saturation, electrodermal activity, skin temperature, sleep duration, BP, and ECG	Besides activity, most parameters that require photoplethysmography are measured only among immobile patients or at rest for ambulatory patients, especially while sleeping	Early detection of physiological response to decompensation	No clearance for most devices and parameters; exceptions for ECG for multiple manufacturers plus irregular rhythm detection; and one device with clearance for BP, pulse, respiratory rate, temperature, and oxygen saturation	Wear device as frequently as possible; charge the device as needed; typically requires Bluetooth connection to a smartphone or other internet-connected device; and be aware of limitations of non-validated data streams	Ability to track continuously, when necessary, or ingest summary statistics periodically; recognise and react to individual deviations from expected norms; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
Patch sensors	Activity, ECG, heart rate variability, respiratory rate, oxygen saturation, skin and derived core temperature, and sleep duration, and quality	For patches with continuous ECG, all vital signs derived from it, including respiration, are available continuously; for non-ECG patches, vital signs data are available only for immobile patients or at rest for ambulatory patients	Early detection of physiological response to decompensation	Majority with clearance for all measured vital signs and some individually derived digital biomarkers	Wear device continuously as long as indicated; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed; and charge or change the device as needed	Ability to track continuously; recognise and react to individual deviations from expected norms; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
Finger pulse oximeters	Oxygen saturation and pulse rate	Primarily recurring spot checks, but continuous is also available	Specific monitoring for hypoxia in settings of infections or exacerbations	Majority with clearance	Use device as indicated or as needed; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed and charge as needed	Ability to capture spot-check data; recognise and react to individual deviations from expected norms; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns (Table 1 continues on next page)

	Parameters measured*	Continuous or intermittent tracking	Potential clinical use	Device FDA regulatory clearance†	User's role‡	Clinician's or health-care system's role‡
(Continued from previous page)						
Thermometers	Body temperature	Primarily spot checks, but also available as a continuous patch sensor	Identification of fever or concerning temperature trajectories suggesting infection	Although considered a class II device, clearance is currently not required	Use device as indicated or as needed; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed and charge as needed	Ability to capture spot-check data; recognise and react to individual deviations from expected norms; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
Home spirometry	Peak expiratory flow, forced expiratory volume in 1 s, and forced vital capacity	Intermittent spot checks	Tracking lung function over time in relation to therapeutic or environment changes	Majority of currently available devices in the USA have clearance	Use device as indicated or as needed; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed and charge as needed	Ability to capture spot-check data; recognise and react to individual deviations from expected norms; seamless incorporation of patient data into electronic health record; and ability to respond to user questions or concerns
Cough frequency detectors	Cough frequency	Continuous	Potential for early detection of decompensation and track clinical course	Most available technologies, primarily apps, are not cleared; one patch sensor has approval for cough detection	Wear approved device continuously as long as indicated; be aware of limitations of non-validated sensor data streams; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed; and charge or change the device as needed	Ability to track cough frequency; recognise and react to individual deviations from expected norms; seamless incorporation of data into electronic health record; and ability to respond to user questions or concerns
Breath analysis	Fractional exhaled nitric oxide and exploratory breath biomarkers	Intermittent spot checks	Identifying and possibly monitoring airway inflammation in asthma	Multiple cleared devices for fractional exhaled nitric oxide, other devices for research only	Use device as indicated or as needed; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed, and charge as needed	Ability to capture spot-check data; recognise and react to individual deviations from expected norm; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
Lung acoustics	Lung sounds	Patches for continuous monitoring and digital personal devices for monitoring lung sounds	Identifying presence and tracking changes in pathological lung sounds	One cleared patch device and multiple cleared digital self-use stethoscopes to transmit breath sounds	Use device as indicated or as needed; maintain Bluetooth connection to a smartphone or other internet-connected device as frequently as needed and charge as needed	Ability to capture spot-check data; recognise and react to individual deviations from expected normal; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
Passive sleep-based monitors	Respiratory rate, pulse rate, and sleep duration and quality	In-bed sensors and near-bed radar systems; continuous tracking when in bed	Early detection of physiological response to decompensation and overall effect of condition on sleep	None currently FDA-cleared	Use device as frequently as possible; maintain Bluetooth connection to a smartphone or other internet-connected device; be aware of limitations of non-validated data streams	Ability to track continuously, when necessary, or ingest summary statistics periodically; recognise and react to individual deviations from expected normal; seamless incorporation of pertinent data into electronic health record; and ability to respond to user questions or concerns
BP=blood pressure. ECG= electrocardiogram. FDA=US Food and Drug Administration. *Device types listed vary regarding the level of validation data supporting their accuracy in measuring the listed parameters. †FDA clearance refers to the FDA allowing a device to market through the 510(k) process based on substantial equivalence to a legally marketed predicate device. No respiratory digital devices have been granted de-novo designation or have been approved via the pre-market assessment pathway. ‡Clinicians and health-care systems will need to educate patients to recognise alarm conditions that require a response; in turn, clinicians and health-care systems will need to respond to increased patient queries regarding alarm conditions.						
Table 1: Personal physiological monitoring technologies available to address challenges in respiratory care						

manner. Considering that an estimated 75% of people who use metered-dose inhalers use them incorrectly or inconsistently, smart inhalers have the potential to substantially improve respiratory illness care.²⁰ Current capabilities of smart inhaler systems, beyond recording exactly when an inhaler was used, include next dose reminders and feedback on technique of proper use of

the inhaler. That information can also be securely shared with the health-care provider. Studies of smart inhalers have shown they can improve therapy adherence, although primarily only when linked with feedback and reminders.²¹

High outdoor air pollution, even when well below regulatory limits, is associated with increased risk of

asthma and COPD exacerbations.^{22,23} Although data from stationary outdoor units can provide a general index of local air pollution, they do not capture individual pollutant and allergen exposures that are dependent on work, school, or home environments. Indoor exposure to particulate matter and increased nitrogen dioxide concentrations also increase exacerbation of respiratory illnesses.^{24,25} A number of indoor air quality sensors, both stationary and wearable, are available commercially and could provide precise exposure histories and real-time information with increased precision.²⁶ Early studies have found that high-efficiency particulate air filters can reduce personal exposure to fine particulate matter of less than 2.5 µm in diameter by about 50%, supporting the potential value of monitoring indoor air quality given that there exists a practical solution to address elevated particulate concentrations.²⁷

A combination of two or more of the technologies mentioned, as in Lynnae's case, can enable a closed-loop system that fosters personal empowerment of the individual with respiratory disease. For example, in the US AIR Louisville programme, the combination of smart inhaler use tracking and monitoring of local outdoor air pollution led to a 78% reduction in rescue inhaler use and a 48% improvement in symptom-free days.²⁸ The development of new systems of care, built around a range of technologies to create personalised, closed-loop solutions for each individual, and supported by continuous tiered connectivity and on-demand educational content, has potential to improve outcomes, lessen the burden of care activities, minimise exacerbations and improve quality of life.

Early identification of acute respiratory illness

As another case example, let us consider Lynnae's fictional husband, Leonard. Leonard had volunteered to participate in a trial that used sensor data from a personal, consumer smartwatch to evaluate its potential value in the early identification of respiratory infection. Lynnae's husband, Leonard, through his participation in a DETECT clinical study, was alerted to early signs of a potential acute respiratory infection based on changes from his normal baseline resting heart rate, movement, and sleep recorded by a commercial wearable device. He was prompted to get a PCR test, which came back positive, leading him to quarantine to avoid exposing Lynnae to COVID-19 and exacerbating her asthma.

Leonard's use of DHTs was key to early detection and corrective action. His viral illness diagnosis was possible through knowledge of his usual physiological response during routine daily activities rather than waiting for his vital signs to fall outside of the range of the population norm (eg, respiratory rate of more than 20 beats per min or temperature of more than 38°C; figure 3). Early data from the USA and other countries, gained over the last three years through COVID-19-related research programmes, such as the one Leonard was participating

in, support the potential value of identifying subtle deviations in a person's physiology and behaviours as an early indicator of an acute viral illness, in some cases even before any noticeable symptoms.^{29–32}

Although the COVID-19 pandemic rapidly accelerated the implementation of DHTs in research programmes, it also surfaced some important challenges related to the use of this technology. For example, coincident with the COVID-19 pandemic, several popular consumer smart watch companies released new functionality to monitor blood oxygen saturation. These functionalities were released as wellness rather than health-care tools, with no evidence of their accuracy reported to, or examined by, regulatory agencies such as the US Food and Drug Administration (FDA).³³ Particularly confusingly, several of these devices have other functionalities that are FDA-regulated (eg, irregular heart rhythm detection), making it difficult for patients and practitioners to understand how to interpret the blood oxygen saturation or other biometric information reported.

For the DETECT study see
<https://detect.scripps.edu>

For the CovidIdentify study see
<https://covididentify.covid19.duke.edu>

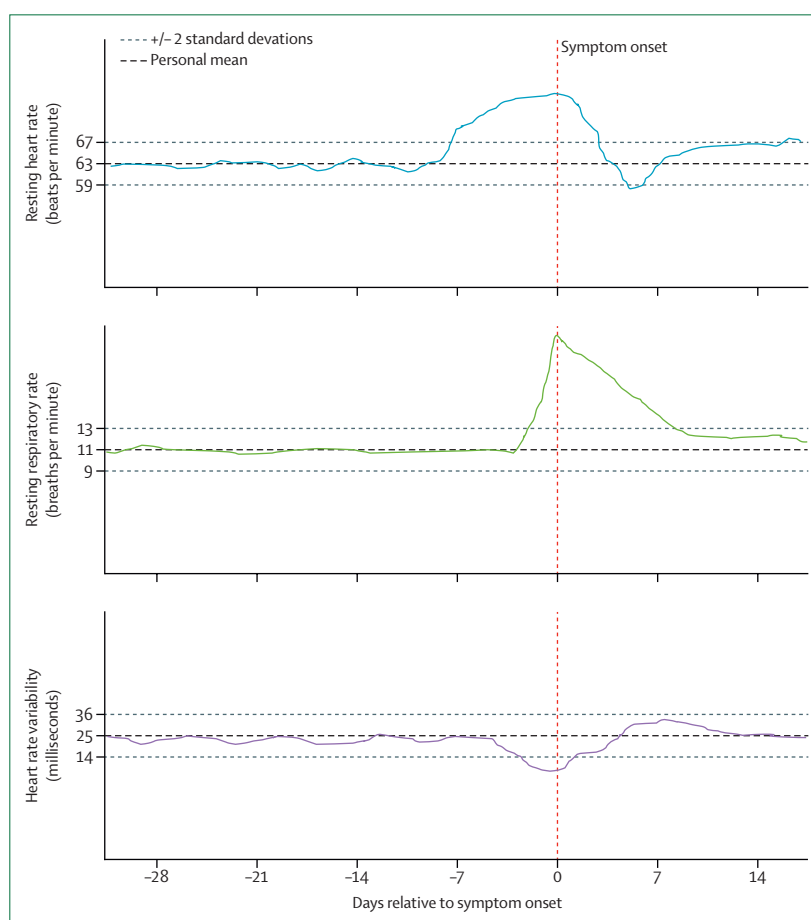


Figure 3: Representative changes in individual physiological parameters preceding symptom onset in individuals with acute COVID-19 infection.

A number of studies, such as CovidIdentify and DETECT, have shown that subtle but individually detectable changes in multiple physiological parameters and behaviours recorded via wearable sensors are associated with acute COVID-19 infection, occasionally preceding the presence of any symptoms.

	Registered on ClinicalTrials.gov			Indexed in PubMed		
	Number of studies	Studies identified with search terms "digital health technologies" and "digital health", "mobile health", or "home"	Studies identified with search terms "digital health technologies" and "pragmatic", "real-world", or "real world"	Overall analytical validation studies	Overall clinical validation studies	Clinical validation studies in respiratory conditions
Auditory diagnostics	535	58	14	39	9	8
Breath	7829	889	82	7	2	2
Cough	4920	542	78	27	7	5
Spirometry	3276	430	46	45	11	9
Pulse oximetry	1658	141	10	77	16	2

Table 2: Studies showing the number and type of validation studies for each type of digital health technology-based measure

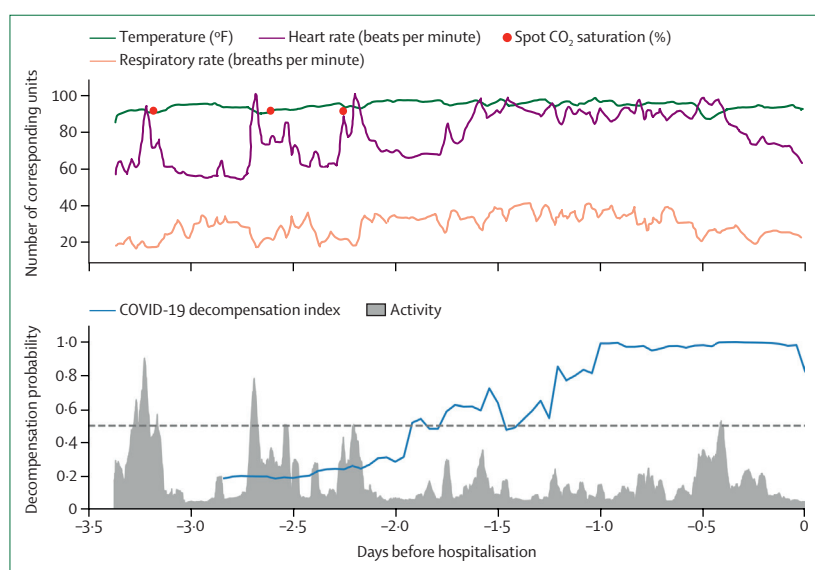


Figure 4: Detection of early indicators of decompensation optimised through machine learning with continuous data streams from a multiparametric sensor patch in individuals newly diagnosed with acute COVID-19
The COVID-19 decompensation index model, based on the unique individual interactions between vital signs and behaviours, has the potential to detect early signs of decompensation relative to any single vital sign. Figure developed using graphics from a paper by Goodday and colleagues⁴³

The need to better address existing health disparities

Another challenge highlighted by many studies including wearables during the COVID-19 pandemic was the reliance on a bring-your-own-device approach, in which only individuals who already owned a wearable device could participate. The major benefit of this design was that it rapidly accelerated the collection of data from many participants without typical geographical barriers. However, because owners of wearable devices are not representative of the general population, results of these studies would more likely contribute to, rather than reduce, existing health disparities.³⁴ As a specific example, optical sensors to measure blood oxygen saturation might not work for people with darker skin tones and for people with the sickle cell trait^{35,36}—both characteristics common in Black populations. Therefore, there is a clear need for digital health studies that sample from populations representative of the target population in

which the sensor is intended to be used. For example, a 2021 small study focused on the use of a connected smart inhaler intervention in children enrolled in Medicaid with asthma in southwest Detroit (MI, USA) found a significant reduction in rescue inhaler use over a mean of over eight months of follow up.³⁷

The need for pragmatic clinical effectiveness trials

FDA clearance of DHTs in the USA reflects that the technology in question measures the variables as described in their labelling. The relation of DHTs to clinical outcomes and value in patient management, however, require additional study. Although gradually increasing, the number of clinical validation studies of DHTs in monitoring respiratory conditions is quite small, with few having evaluated clinical effectiveness in improving health quality in real-world settings (table 2). Although effectiveness trials are necessary and require broad support going forward, the current scarcity of such studies should not diminish enthusiasm for their potential, especially considering that the certainty of evidence supporting the majority of guideline-recommended therapies in the 2020 Asthma Guideline update was low to moderate.³⁸

Changing roles for clinicians and patients

Digital health data are unique in their continuous and longitudinal format, which is unfamiliar to most patients and clinicians and will require both parties to take on new roles during management (table 1). Digital health implementation will require increased patient agency and empowerment in their own care; thus, the patient will need to recognise alarms from devices and clinicians will need to respond to increasing patient concerns about these deviations from a person's baseline. However, with primary care providers already overburdened, requiring more hours than there are in the day to just provide guideline-directed primary care, expecting that digital health data can be broadly incorporated into current systems of care in a meaningful way, is unrealistic.³⁹ To address this obstacle to innovation, several health systems have developed dedicated digital health programmes,⁴⁰ and a growing number of virtual care organisations have been developed, with at least one specifically focused on

people with respiratory conditions built around remote monitoring and virtual care.⁴¹

Data might come in the form of a single measurement per day or per week (ie, daily or weekly summaries), or might be higher frequency, up to the resolution of one or several measurements per minute or per second. Given the newness of this data type in clinical practice, analysis and interpretation challenges remain. One solution has been the development of digital biomarkers that serve to transform high-volume digital health technology data (eg, minute-by-minute measurements of blood oxygen saturation from a pulse oximeter), into indicators of health outcomes (eg, COPD exacerbation risk). The digital biomarker can convert large streams of multivariate data into a single indicator that is useful as a threshold for clinical decision making and to lend context (eg, if a patient is borderline *vs* an extreme case). One example of this was the development of a COVID-19 decompensation index using wearable sensor data to alert the care team if an individual was at risk for decompensation requiring hospitalisation following an acute COVID-19 diagnosis (figure 4).⁴²

Beyond the practical challenges of implementation are the emotional ones. Digitally enabled systems of care can disrupt the traditional power balance between patients and physicians.⁴³ Digitally-enabled systems participatory medical models can return agency to patients, allowing them to manage their condition in partnership with their provider, and even on their own in an informed manner supported by individualised data. Given this disruption to a power dynamic established over centuries, this transition will neither be immediate nor easy for patients or providers.

Digitally-enabled systems of care is still very much a nascent field that is rapidly evolving to meet the practical needs of clinicians and patients. Cross-disciplinary collaborations bridging across the different areas required for digital biomarker development (ie, statistical, mathematical, computational, and respiratory health expertise) are needed. Moreover, tools that allow for collaboration in improving data-driven algorithms possible with DHTs, validating known digital biomarkers, and discovering new digital biomarkers will enable much-needed standardisation and interoperability in this burgeoning field.⁴⁴ Finally, implementation scientists and human factors engineers will need to assess the user experiences and areas where human error might obfuscate the use of a digital biomarker in practice. Other aspects to be elucidated include the legal consequences of the medical applications of DHTs and the inclusion and evolution of artificial intelligence in such applications.

Summary and future expectations

As a leading cause of morbidity and mortality globally, respiratory diseases present a substantial management challenge and highlight the vulnerability of all people to environmental and pathogen triggers of decompensation.

Search strategy and selection criteria

References for this Viewpoint were identified through a search of PubMed and Google Scholar for articles published from Jan 1, 2010, to Sept 30, 2023, by use of the terms “respiratory”, “pulmonary”, “COPD”, “asthma”, “digital health”, “wearable”, and “sensor”. Other relevant references were identified from key online sources, including consumer health meetings by JP Morgan Consumer Electronics Show, and authors’ personal files. Only articles published in English were included. Table 2 is not the result of a systematic review. Rather, our search was done as follows: clinical trials: ClinicalTrials.gov was searched on Nov 13, 2023, with the use of search terms listed in the column headers, where digital health technologies (DHTs) in columns 3 and 4 corresponds to the row name in column 1. For example, the data entry in column 3 row 1 uses the search terms auditory diagnostics and (“digital health”, “mobile health”, or “home”). PubMed publications: these numbers come from a November, 2023 search of HumanFirst’s Atlas database, a comprehensive library of studies using digital measures and technologies. Only data from publications in English were included in the analysis. Publications are flagged for review based on curated search queries aiming to surface DHT studies. Each study is then manually reviewed for the clinical population(s) evaluated, the digital measures captured, and the technologies used to capture them. If all can be identified, including the specific make and model of technology, it is included in the Atlas library.

Reducing the global burden of respiratory illness can be aided by taking advantage of a range of transformational DHTs. At the individual level, continuous measures allow for analytics comparing the present moment to well established historical data and alerting the patient and their health-care providers that they are moving out of their normal operating zone, which will allow for changes in management. The prospects for avoiding emergency room visits and intensive care unit stays are good but there is evidence that underserved populations do not have access to DHTs to benefit similarly to those who are well-served by health care. Summing digital data across communities will probably provide public health officials with the means to monitor emerging outbreaks and pandemics and help reduce morbidity and mortality from emerging environmental exposures due to climate and industry. Lynnae was lucky that she, her family, and her health-care providers were early adopters of DHTs—now for the rest of the world to follow.

Contributors

All authors conceptualised the manuscript, researched its contents, wrote the manuscript, and edited all revisions. All authors have read and approved the final manuscript.

Declaration of interests

JD reports personal fees as a Scientific Advisor to Veri and grant support from AstraZeneca. AC is the CEO, co-founder, and a shareholder of HumanFirst; and was a board member of the Digital Medical Society until 2023. MF is an advisor and shareholder of HumanFirst. SRS reports financial compensation as a consultant for physiIQ; declares research grant support from Janssen Research & Development; and received reimbursement as an Executive Committee member for the Heartline Study, which is a trial on wearables to detect atrial fibrillation. GSG declared no competing interests.

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